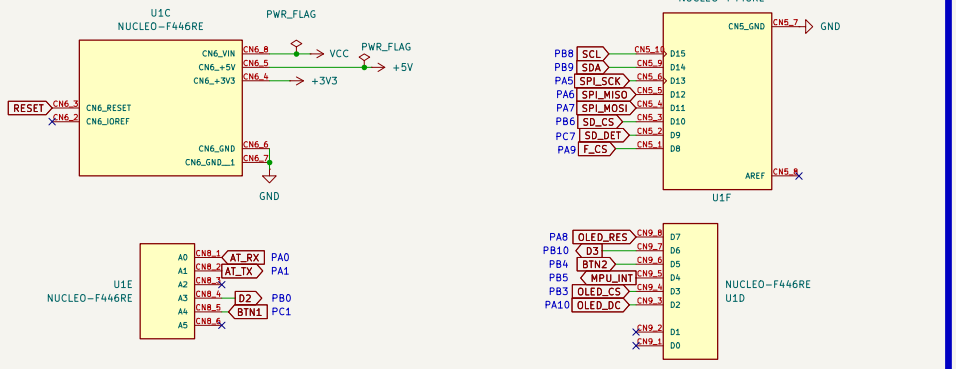
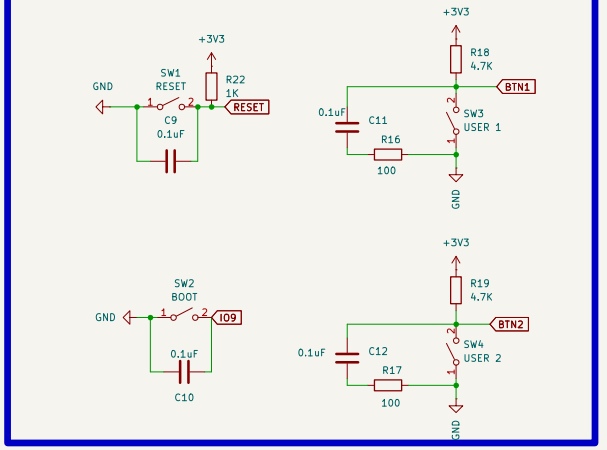
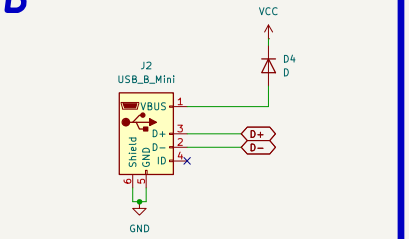
HEADER PINS



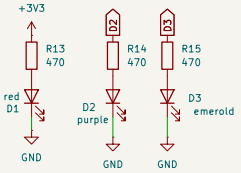
BUTTONS



USB



LEDS



SENSORS

File: SENSORS.kicad_sch

ESP32C6

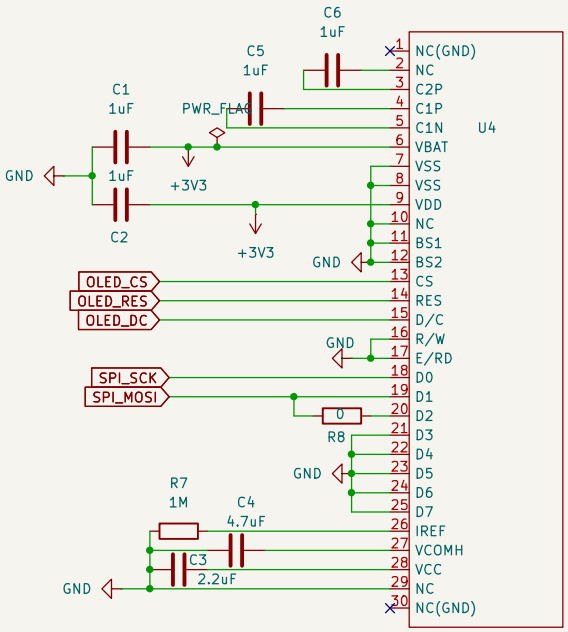
File: ESP32C6.kicad_sch

OLED

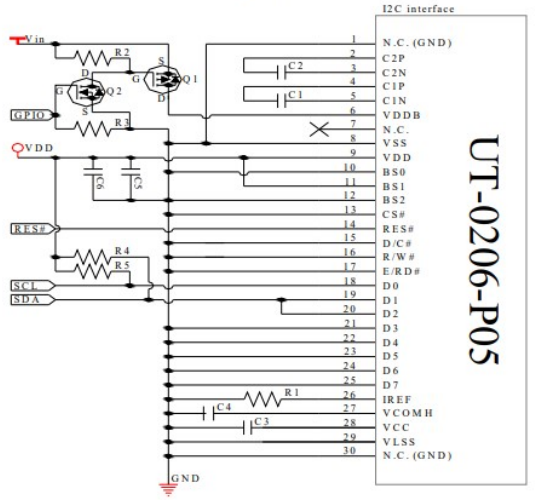
File: OLED.kicad_sch

uSD_slot

File: uSD_slot.kicad_sch



3.3.5.2 I²C Interface with Internal Charge Pump



Recommended Components:

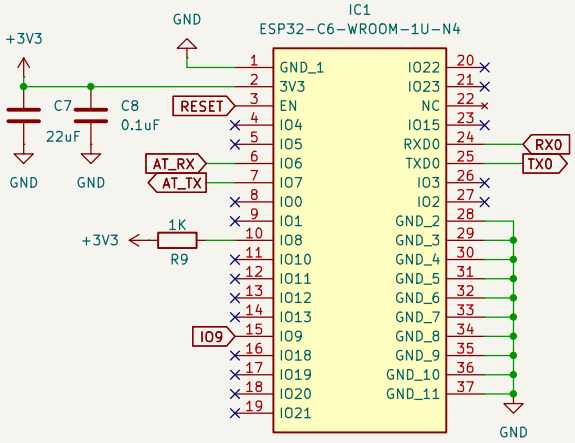
C1, C2:	1μF / 16V, X5R
C3:	2.2μF
C4:	4.7μF / 16V, X7R
C5, C6:	1μF
R1:	560kΩ, R1 = (Voltage at IREF - VSS) / IREF
R2, R3:	47kΩ
R4, R5:	4. 7kΩ
Q1:	FDN338P
Q2:	FDN335N

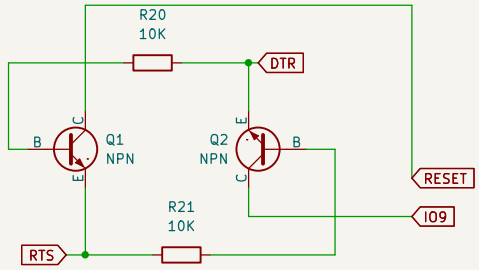
Notes:

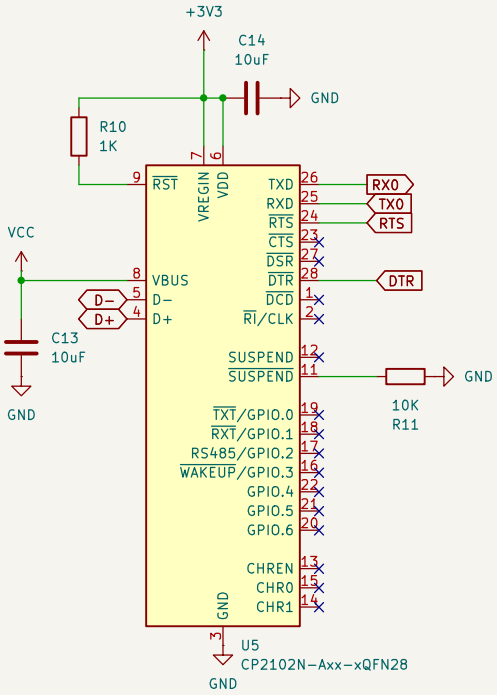
VDD:	1.65~3.3V, it should be equal to MPU I/O voltage.
Vin:	3.5~4.2V

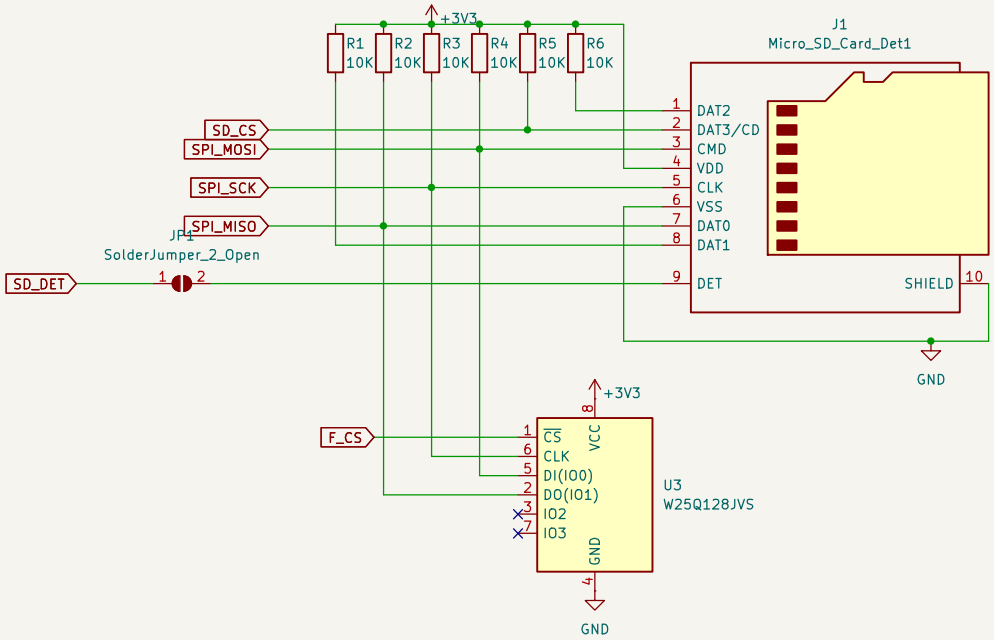
The I²C slave address is 0111100b'. If the customer ties D/C# (pin 15) to VDD, the I²C slave address will be 0111101b'.

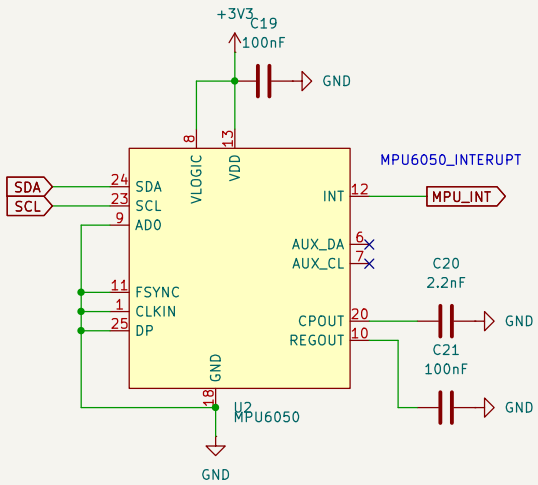
* VBAT will be connected to VDD when VCC be connected to external source (12V), R1 should be replaced as **910 kΩ**.

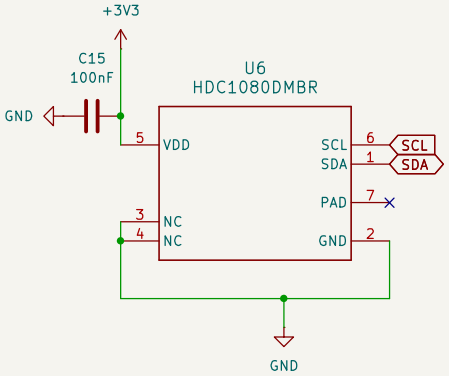

IC1
ESP32-C6-WROOM-1U-N4
GND
+3V3
C7
22uF
C8
0.1uF
RESET
AT_RX
AT_TX
+3V3
1K
R9
IO9
GND_1
3V3
EN
IO4
IO5
IO6
IO7
IO0
IO1
IO8
IO10
IO11
IO12
IO13
IO9
IO18
IO19
IO20
IO21
IO22
IO23
NC
IO15
RXD0
TXD0
IO3
IO2
GND_2
GND_3
GND_4
GND_5
GND_6
GND_7
GND_8
GND_9
GND_10
GND_11
RX0
TX0
GND


R20
10K
DTR
Q1
NPN
Q2
NPN
B
C
E
RESET
R21
10K
IO9
RTS


+3V3
C14
10uF
GND
R10
1K
VCC
C13
10uF
GND
RST
VREGIN
VDD
VBUS
D-
D+
TXD
RXD
RTS
CTS
DSR
DTR
DCD
RI/CLK
SUSPEND
SUSPEND
TXT/GPIO.0
RXT/GPIO.1
RS485/GPIO.2
WAKEUP/GPIO.3
GPIO.4
GPIO.5
GPIO.6
CHREN
CHR0
CHR1
GND
RX0
TX0
RTS
DTR
10K
R11
U5
CP2102N-Axx-xQFN28
GND


+3V3
R1
10K
R2
10K
R3
10K
R4
10K
R5
10K
R6
10K
J1
Micro_SD_Card_Det1
SD_CS
SPI_MOSI
SPI_SCK
SPI_MISO
JP1
SolderJumper_2_Open
SD_DET
1
2
DAT2
DAT3/CD
CMD
VDD
CLK
VSS
DAT0
DAT1
DET
SHIELD
10
GND
F_CS
+3V3
CS
CLK
DI(IO0)
DO(IO1)
IO2
IO3
VCC
GND
U3
W25Q128JVS
GND


+3V3
C19
100nF
GND
SDA
SCL
24
23
9
SDA
SCL
ADO
8
VLOGIC
13
VDD
MPU6050_INTERUPT
INT
12
MPU_INT
AUX_DA
AUX_CL
6
7
C20
2.2nF
11
1
25
FSYNC
CLKIN
DP
CPOUT
REGOUT
20
10
GND
C21
100nF
GND
18
GND
U2
MPU6050
GND


+3V3
C15
100nF
GND
U6
HDC1080DMBR
5
VDD
SCL
SDA
6
1
SCL
SDA
PAD
7
3
4
NC
NC
GND
2
GND